



SGGS Institute of Engineering and Technology, Vishnupuri, Nanded

End Term (Semester-I) 2025-26

Final Year. B. Tech.

Mechanical Engineering Department

Renewable Energy

PEC-ME414

Date: 18/11/25

Time: 13:30 to 15:30

Max. Marks :60

- Note:**
- All questions are compulsory.
 - Use of non-programmable calculator is permitted.
 - This paper contains ONE page.

Q.N.		Question	Marks	CO	BT
Q.1	(a)	What are renewable energy resources? Discuss their importance in India's power requirement context	8	1	2
	(b)	How does environment get affected by the use Solar energy and wind energy resource	7	1	3
Q.2	(a)	Describe the construction and working of a solar distillation. How the efficiency of solar distillation can be increased.	8	2	3
	(b)	Explain working of Solar dryer with neat sketch and its types.	7	2	2
Q.3	(a)	Explain the classification of wind turbines and the basic components of a horizontal axis wind turbine.	8	3	2
	(b)	Wind at one standard atmospheric pressure and 15°C has a speed of 10 m/s. A 10-m diameter wind turbine operates with maximum efficiency of 40%. Calculate (i) the total power density in wind stream, (ii) the maximum power density, (iii) the actual power density, (iv) the power output of the turbine.	7	3	4
Q.4	(a)	Differentiate between the following types of electric motors used in Electric Vehicles (EVs). Write any five points of comparison for each motor: Brushed DC Motor (BDC) Brushless DC Motor (BLDC) Induction Motor (IM) Permanent Magnet Synchronous Motor (PMSM) Switched Reluctance Motor (SRM)	8	4	3
	(b)	Differentiate between the following five types of fuel cells. Write five points of comparison for each type: Proton Exchange Membrane Fuel Cell (PEMFC) Alkaline Fuel Cell (AFC) Phosphoric Acid Fuel Cell (PAFC) Molten Carbonate Fuel Cell (MCFC) Solid Oxide Fuel Cell (SOFC)	7	4	3

Course Outcomes (CO)

- CO 1: Interpret energy reserves of India and potential of different energy sources.
CO 2: Measure the solar radiation parameters and performance of different solar collectors.
CO 3: Calculate different parameters of wind turbine rotor.
CO 4: Demonstrate knowledge in field of fuel cell and potential for power generation

Bloom's Taxonomy (BT)

BT1- Remember, BT2- Understand, BT3- Apply, BT4- Analyze, BT5- Evaluate, BT6- Create



S. G. G. S. Institute of Engineering and Technology, Vishnupuri, Nanded.
End Sem Examination (Sem I) 2025-26
Class: Final Year (Mechanical Engineering)

Date: 18/11/2025
Time: 13:30PM To 15:30PM

Subject: Quality & Reliability Engineering (PCC-ME412)

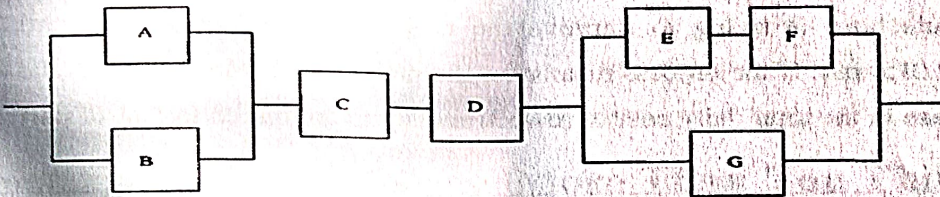
Max. Marks: 60

Use of non-programmable calculator is permitted.

Assume suitable data if necessary.

Stat tables (A1 to A7) and formulae of control charts with my signature are allowed for examination.

Q.1 to Q.4 are mandatory. Attempt any four questions from Q.5 to Q.10.

Sr. No.	Questions	M	CO	BT
Q.1.	Consider the seven-component system shown in Figure 1.1.1. The reliabilities of the components are as follows: $R_A = 0.96$, $R_B = 0.92$, $R_C = 0.94$, $R_D = 0.89$, $R_E = 0.95$, $R_F = 0.88$, $R_G = 0.90$. Find the reliability of the system. If you had a choice of improving system reliability by modifying any two components, how would you proceed? a) Find the trial control limits and construct a respective chart. b) Assuming the thickness of the coating to be normally distributed, what proportion of the product will not meet specifications?	(10)	CO6	3
				
Consider the seven-component system shown in above figure. Assume that the time to failure for each component has an exponential distribution. The failure rates are as follows: $\lambda_A = 0.0005/\text{hour}$, $\lambda_B = 0.0005/\text{hour}$, $\lambda_C = 0.0003/\text{h}$, $\lambda_D = 0.0008/\text{hour}$, $\lambda_E = 0.0004/\text{hour}$, $\lambda_F = 0.006/\text{hour}$, and $\lambda_G = 0.0064/\text{hour}$. Find the reliability of the system after 1000 hours. What is the mean time to failure of the system?				
Q.2.	Find a single sampling plan that satisfies a producer's risk of 5% for lots that are 1.8% nonconforming, and a consumer's risk of 10% for lots that are 9% nonconforming.	(10)	CO5	4
Q.3.	The number of dietary errors is found from a random sample of 100 trays chosen on a daily basis in a health care facility. The data for 25 such samples are shown in following table. (a) Construct an appropriate control chart and comment on the process. (b) How many dietary errors do you predict if no changes are made in the process? (c) Is the system capable of reducing dietary errors to 2, on average, per 100 trays, if no changes are made in the process?	(10)	CO4	3

Sample Number	Number of Dietary Errors	Sample Number	Number of Dietary Errors
1	9	14	8
2	6	15	8
3	4	16	7
4	7	17	6
5	5	18	4
6	6	19	12
7	16	20	7
8	8	21	6
9	7	22	8
10	9	23	6
11	3	24	8
12	6	25	5
13	10		

- Q.4.** The specifications for the thickness of nonferrous washers are 1.0 ± 0.04 mm. From process data, the distribution of the washer thickness is estimated to be normal with a mean of 0.98 mm and a standard deviation of 0.02 mm. The unit cost of rework is \$0.10, and the unit cost of scrap is \$0.15. For a daily production of 10,000 items: (10) CO1 3
- (a) What proportion of the washers is conforming? What is the total daily cost of rework and scrap?
- (b) The manufacturer is trying to improve the process and reduce its standard deviation to 0.015 mm. If the process means are maintained at 1.0 mm, what is the percent decrease in the total daily cost of rework and scrap compared to that of part (a)?
- Q.5.** a) Find the reliability of a 3-unit parallel system with the reliability of units as 0.6, 0.7, and 0.8. (05) CO6 3
- b) For the above parallel system, should we add one more unit with reliability of 0.6, or improve the reliability of each unit by 10% to improve the reliability of the system to a great extent?
- Q.6.** The reliability of a hydraulic press is determined to be 0.88 for a designed operating time of 10,000 hours. A failure of the hydraulic press will cost Rs. 28,000. The reliability of press can be increased to 0.96 by 20 hours of preventive maintenance at a cost of Rs. 65/- per hour. Should preventive maintenance be performed? (05) CO6 3
- Q.7.** Distinguish between producer's risk and consumer's risk. In this context, explain the terms acceptable quality level and limiting quality level. Discuss instances for which one type of risk might be more important than the other. (05) CO5 2
- Q.8.** Let's consider a double sampling plan of lot size 3000 given by the following parameters: $n_1 = 40$, $c_1 = 1$, $r_1 = 5$, $n_2 = 80$, $c_2 = 5$, $r_2 = 6$. For a lot of proportion nonconforming value of $p = 0.03$, find the probability of accepting such lots. (05) CO5 3

Q.9. The development of new drugs has to undergo stringent regulations resulting in long lead times and high research and development costs. Suppose that you are in charge of the federal agency that grants drug approvals, and you wish to be convinced that a new drug is better than an existing one in terms of increasing the average life of patients. (05) CO2 2

- (a) What are appropriate null and alternative hypotheses?
 (b) Explain what type I and II errors are in this context, and discuss their relative seriousness.

Q.10. A pharmaceutical company producing vitamin capsules desires a proportion of calcium content between 40 and 55 ppm. A random sample of 20 capsules chosen from the output yields a sample mean calcium content of 44 ppm with a standard deviation of 3 ppm. find the C_p index. Comment on the ability of the process to meet specifications. What proportion of the specification range is used up by the process? (05) CO4 3

Course Outcomes:

- CO 1 Interpret the basic principles of probability theory, standard distributions.
 CO 2 Understand and solve numerical on confidence interval, test of hypothesis.
 CO 3 Solve numerical examples on statistical process control, rules for out of control, average run length, warning limits, probability of false alarm
 CO 4 Solve numerical examples on variable and attribute control charts, process capability analysis.
 CO 5 Design a single/double sampling plan, construct its OC curve, and estimate ATI, AOQ and other indices, evaluate economics of sampling.
 CO 6 Solve numerical on system reliability estimation.

Bloom's Taxonomy (BT):

BT1- Remember, BT2- Understand, BT3- Apply, BT4- Analyze, BT5- Evaluate, BT6- Create

~ALL THE BEST~

“Trust Yourself, You Know More Than You Think You Do”

TABLE 10-5 Values of np for a Producer's Risk of 0.05 and a Consumer's Risk of 0.10

Acceptance Number, c	$P_a = 0.95, np_1$	$P_a = 0.10, np_2$	np_2/np_1
0	0.051	2.303	44.84
1	0.355	3.890	10.96
2	0.818	5.322	6.51
3	1.366	6.681	4.89
4	1.970	7.994	4.06
5	2.613	9.274	3.55
6	3.286	10.532	3.21
7	3.981	11.771	2.96
8	4.695	12.995	2.77
9	5.426	14.206	2.62
10	6.169	15.407	2.50
11	6.924	16.598	2.40
12	7.690	17.782	2.31
13	8.464	18.958	2.24
14	9.246	20.128	2.18
15	10.035	21.292	2.21

“Source: F.E. Grubbs, “On Designing Single Sampling Plans,” *Annals of Mathematical Statistics*, XX: 256, 1949. Reprinted by permission of the Institute of Mathematical Statistics.



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Mechanical Engineering
End Semester Examination
Academic Year: 2025-26 (Semester – I)

Class: Final Year B.Tech.	Subject: Introduction to Industry 4.0 (Elective - VI)	Subject Code: PEC-ME-404
Date: 22.11.2025	Time: 13.30 to 15.30	Max. Marks: 60

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Mobile phones and programmable calculators are strictly prohibited.
- iii. Exchange/sharing of stationery, calculator, etc. is not allowed.
- iv. Figures to the right indicate full marks.
- v. Indicate the Question Numbers while answering each question correctly.

Q. No.		Solve any TWO Questions out of THREE Questions in EACH.	Marks	CO	BT
Q.1	(a)	Discuss the key features and major inventions of each Industrial Revolution in Industry 4.0.	06	CO1	BT2
	(b)	Explain digitization in Industry 4.0, its connection to the networked economy, and the challenges it brings.	06	CO1	BT2
	(c)	Describe the progress of Industry 4.0 in the USA, Europe, and China. Why has India not yet achieved similar advancements?	06	CO1	BT2
Q.2	(a)	Explain the Internet of Things (IoT) and its role in smart manufacturing with examples.	06	CO2	BT2
	(b)	Differentiate between IoT and IIoT. What are the key features of IIoT that make it suitable for industrial applications?	06	CO2	BT4
	(c)	Explain Smart Manufacturing. Describe its main features and benefits with a suitable example.	06	CO2	BT2
Q.3	(a)	Describe the concept of Cyber-Physical Systems with the help of a standard block diagram.	06	CO3	BT2
	(b)	Compare traditional industrial robots with collaborative robots, citing at least three key differences in functionality and application.	06	CO3	BT4
	(c)	Outline the fundamentals of cloud computing. Discuss its specific applications in enabling key aspects of Industry 4.0, such as smart factories, IoT integration, and big data processing.	06	CO3	BT1
Q.4	(a)	Explain how data, information, knowledge, and collaboration work together to support digital transformation in Industry 4.0.	06	CO4	BT2
	(b)	Explain the Resource-Based View (RBV) and how it helps companies use their resources for competitive advantage.	06	CO4	BT2
	(c)	Explain the benefits of cloud computing for businesses in terms of cost savings, accessibility, and collaboration.	06	CO4	BT2
Q.5	(a)	What are the challenges and opportunities related to system integration in Industry 4.0?	06	CO5	BT1
	(b)	How do Siemens, GE, and Boeing use new Industry 4.0 technologies to improve their businesses?	06	CO5	BT4
	(c)	What skills do workers need to use AI and Industry 4.0 technologies in automation and manufacturing?	06	CO5	BT1

Bloom's Taxonomy (BT)

Course Outcomes:

On successful completion of this course, students should be able to:

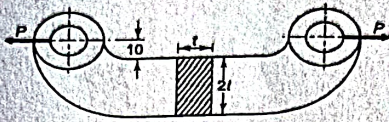
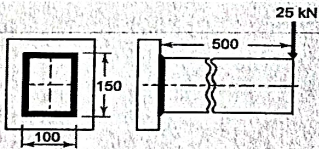
CO 1	Understand the drivers and enablers of Industry 4.0
CO 2	Appreciate the smartness in Smart Factories, Smart cities, smart products, and smart services.
CO 3	Identify the various systems used in a manufacturing plant and their role in an Industry 4.0 world.
CO 4	Appreciate the power of Cloud Computing in a networked economy.
CO 5	Understand the opportunities, challenges brought about by Industry 4.0 and how organizations and individuals should prepare to reap the benefits.

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

	SGGS Institute of Engineering and Technology, Vishnupuri, Nanded		
	Academic Year: 2025-26	Semester: I	Examination: End-term
	Class: Third Year	Branch: Mechanical Engineering	
	Date: 24/11/2025	Time: 10:00 - 12:00	
Subject: Machine Design - I		Code: PCC-ME305	Maximum Marks: 60

General Instructions:

- I. This paper contains 1 (One) page. Figures to right indicate full marks.
- II. Draw neat & compact sketches wherever necessary.
- III. Assume data wherever required and state them clearly.

Q.N.	Question	Marks	CO	BT
1.	State selection criterion for different theories of failure under the static loading.	5	CO2	BT1
2.	An offset link subjected to a force of 25 kN is shown in figure. It is made of grey cast iron FG300 and the factor of safety is 3. Determine the dimensions of the cross-section of the link. (Dimensions in mm.) 	10	CO2	BT5
3.	Explain Endurance stress, Endurance limit along with the <i>S-N</i> curve.	5	CO3	BT2
4.	A rotating bar made of steel 45C8 ($S_{ut} = 630 \text{ N/mm}^2$) is subjected to a completely reversed bending stress. The corrected endurance limit of the bar is 315 N/mm^2 . Calculate the fatigue strength of the bar for a life of 90,000 cycles. [Hinh: Use <i>S-N</i> Curve.]	10	CO3	BT3
5.	Classify the keys along with neat and clean sketches.	5	CO4	BT1
6.	A hollow transmission shaft, having inside diameter 0.6 times the outside diameter, is made of plain carbon steel 40C8 ($S_{yt} = 380 \text{ N/mm}^2$) and the factor of safety is 3. A belt pulley, 1000 mm in diameter, is mounted on the shaft, which overhangs the left hand bearing by 250 mm. The belts are vertical and transmit power to the machine shaft below the pulley. The tension on the tight and slack sides of the belt are 3 kN and 1 kN respectively, while the weight of the pulley is 500 N. The angle of wrap of the belt on the pulley is 180° . Calculate the outside and inside diameters of the shaft. based on the maximum shear stress theory, given as $\tau_{max} = \frac{16}{\pi d_o^3 (1 - C^4)} \sqrt{M_b^2 + M_t^2},$ where the notations has their usual meanings. [Draw explicit sketch]	10	CO4	BT5
7.	A beam of rectangular cross section is welded to a support by means of fillet welds as shown in figure. Determine the size of the welds, if the permissible shear stress in the weld is limited to 75 N/mm^2 . (Dimensions in mm.) 	15	CO5	BT5

**** ALL THE BEST ****

CO1: Analyze the fundamental principles of machine design to identify and evaluate the design requirements, material selection, and safety considerations for various mechanical components.

CO2: Apply theories of static failure to design and analyze mechanical components subjected to static loading.

CO3: Evaluate and design mechanical components to withstand fluctuating and fatigue loading conditions using endurance limit estimation.

CO4: Design and analyze shafts, keys, and couplings as per industrial standards.

CO5: Develop designs for threaded and welded joints subjected to static, eccentric, and fluctuating loads.

BT1: Remember, BT2: Understand, BT3: Apply, BT4: Analyze, BT5: Evaluate, BT6: Create.